

International Scientific and Practical Conference

“Digital Readiness for Big Data Use in Schools: The Case of Uzbekistan”

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Contents

Introduction

Research
Questions

Research
Background

Objectives

Methodology

Data and
Results

Conclusion

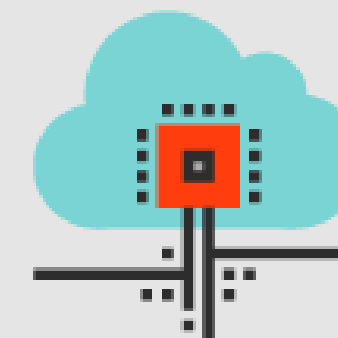
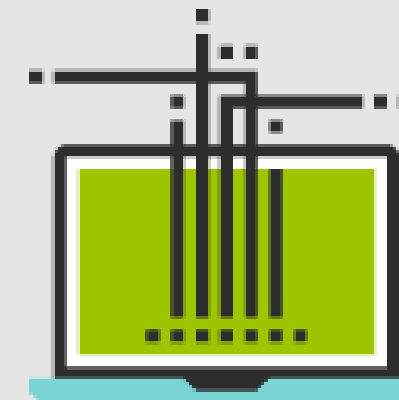
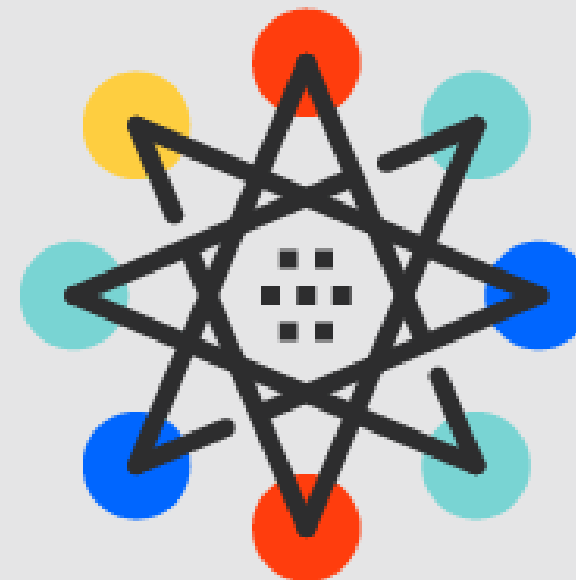
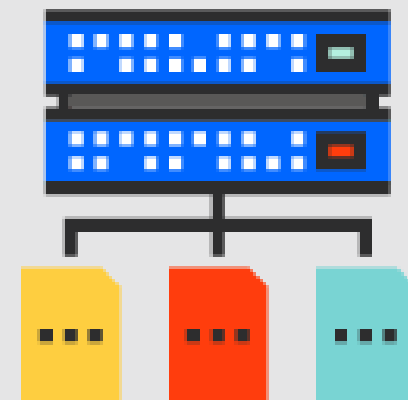
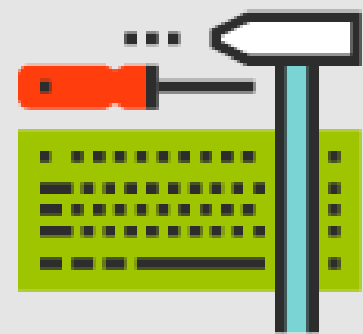
Recommendations

Introduction

In an era where major decisions are made based on data, overlooking the potential of big data in the education system can result in irreparable missed opportunities.



By analyzing questionnaire data and semi-structured interviews, this research will attempt to present a comprehensive picture of the current situation and provide practical suggestions for policymakers, curriculum designers, and administrators of educational institutions.



Research question

1. To what extent do Uzbek school teachers use data, especially big data, in their teaching?
2. What is the level of digital skills among schoolteachers in Uzbekistan?
3. What individual, organizational, and technological factors hinder the effective use of big data in education?
4. What kind of training or support is needed to enhance digital and data-driven capabilities?
5. What solutions can be implemented to develop a data-driven approach in education?

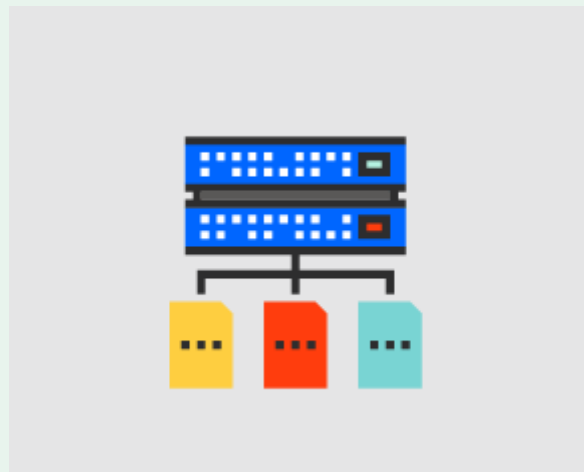
Research background



The Concept and Theoretical Framework of “Big Data” in Education



Challenges and Barriers to Developing Data-Driven Education



Applications and Benefits of Big Data Analytics in Education



Research Gaps and Future Directions

Research objectives



Objective 1

The objectives of this research, in the first stage, are to examine the current use of big data in the educational process in schools and to measure the level of digital skills among school teachers in Uzbekistan.



Objective 2

The next step is to identify the challenges, obstacles, and educational needs for the development of data-driven education in Uzbekistan, as well as to provide practical solutions and suggestions for developing a data-driven approach in education.

Methodology

The analysis was conducted twice using the same set of 170 cross-sectional survey responses, once with digital skills as the dependent variable and once with big data usage as the dependent variable, while keeping the independent variables constant across both models.

Part 4: Challenges and needs

18. Lack of data analysis knowledge and skills is an obstacle to the effective use of big data in education.
19. Technical infrastructure (internet and appropriate information system) is a barrier to data analysis.
20. Managers and policymakers' support for a data-driven approach is not enough.
21. I need more training to make better use of training data.

Part 5: Open Questions

22. Given the right resources and training, what types of data would you be willing to use to improve your teaching?
23. What actions and changes do you think are necessary to better utilize big data in education?

Dear Participant,

This questionnaire is part of an academic research project aimed at exploring the use of data in educational processes, assessing digital skills, and identifying the challenges and needs for developing data-driven education among teachers and university lecturers.

Your input will help us gain a clearer understanding of the current situation and develop practical recommendations for improving digital education. Completing the questionnaire will take approximately 10 minutes. All responses will remain **strictly confidential** and will be used solely for research purposes.

Thank you in advance for your valuable cooperation!

Part 1: Demographic information

1. Gender: Male ☐ Female ☐
2. Age: less than 30 ☐ 30-40 ☐ 41-50 ☐ greater than 50 ☐
3. Education Level: Bachelor ☐ Master ☐ PhD ☐
4. Teaching area: Humanities ☐ Basic Sciences ☐ Technology and Engineering ☐ Social Sciences ☐ Other (specify).....[_____](#)
5. Teaching experience: less than 5 years ☐ 5-10 years ☐ 11-20 ☐ more than 20 ☐
6. Where do you teach? Private school ☐ Governmental school ☐ International School ☐ Others (specify).....[_____](#)

Part 2: Digital Skills

Completely opposite=1 Opposite=2 No Idea=3 Agree=4 Totally agree=5

7. I am skilled in using digital educational software (such as Google Classroom, Zoom, and LMS).
8. I am able to analyze simple student data (either with Excel, graphs or tables).
9. I am familiar with the basic concepts of data security and privacy.
10. I have received specific training in digital skills throughout my professional career.
11. My organization has provided courses to improve teachers' digital literacy.
12. I am interested in learning data analysis tools in education.

Part 3: Using data in education

13. I have a history of using student performance data to improve teaching methods.
14. I use digital systems to collect and analyze educational data.
15. Data-driven information has helped me make educational decisions.
16. In my workplace, there is a culture of using data to make decisions.
17. The use of data has improved learning outcomes.

Data and Results

The regression model was found to be statistically significant, explaining approximately 72% of the variance in digital skills among teachers ($R^2 = 0.72$, Adjusted $R^2 = 0.68$). The overall F-statistic indicated that the model was a good fit for the data ($F = 10.25$, $p < 0.001$).

Digital Skills

Explained Variable	Coefficient	Std.err	t	probability
Gender	0.08	0.10	0.80	0.425
Age	0.01	0.02	0.50	0.615
Education Level*	0.30	0.18	1.67	0.095
Experience**	0.28	0.19	1.47	0.145
Type of School	0.05	0.07	0.71	0.478
Cons	0.95	0.25	3.80	0.001

* Significance for Educational Level is around 81–82%

** Significance for Experience is around 72–78%

$$\text{Digital Skills} = \beta_0 + \beta_1(\text{Gender}) + \beta_2(\text{Age}) + \beta_3(\text{Education Level}) + \beta_4(\text{Experience}) + \beta_5(\text{Type of School}) + \varepsilon$$

Data and Results

The results indicate an R-squared value of 0.68, suggesting that approximately 68% of the variance in big data usage can be explained by the selected independent variables. This demonstrates a relatively good fit of the model.

Big Data

Explained Variable	Coefficient	Std.err	t	probability
Gender	0.10	0.09	1.11	0.270
Age	0.02	0.02	1.25	0.210
Education Level	0.42	0.18	2.33	0.045
Experience	0.38	0.20	1.90	0.065
Type of School	0.07	0.06	1.17	0.245
Cons	1.10	0.20	5.50	0.000

* Significance for Educational Level is around 87–90%

** Significance for Experience is around 84–85%

$$\text{Big Data} = \beta_0 + \beta_1(\text{Gender}) + \beta_2(\text{Age}) + \beta_3(\text{Education Level}) + \beta_4(\text{Experience}) + \beta_5(\text{Type of School}) + \varepsilon$$

Conclusion

1

Teachers' education and professional experience were both positively correlated with digital competency as well as big data use, showing the need for focused capacity-building programs.

2

By synthesizing quantitative and qualitative results, this study contributes to the growing body of literature on data use in education in developing contexts and offers actionable recommendations for policymakers, curriculum designers, and educational leaders.

Recommendations

1. Develop a National Policy for Data–Driven Education
2. Invest in School and Teacher Digital Infrastructure
3. Offer Regular Capacity–Building Programs for Teachers
4. Establish a Data–Informed Organizational Culture
5. Facilitate Local Research and Indigenous Data Production
6. Indigenous Learning Analytics Platforms
7. Ethical Considerations and Data Privacy
8. Establish Allocated Budgets for Education Digital Transformation

WITH APPRECIATION FOR

YOUR KIND ATTENTION

THANK YOU!



Any questions?